Extrema and the Mean Value Theorem

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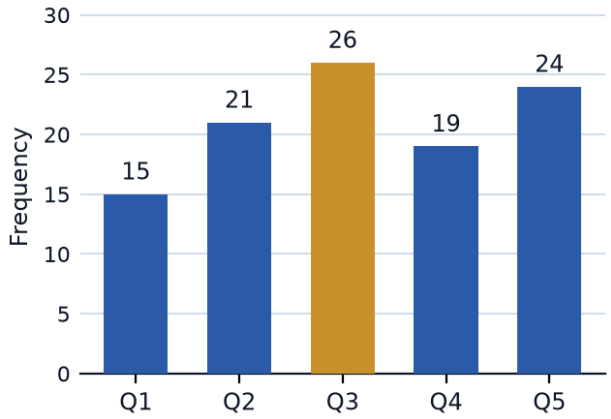
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Critical points and extrema

Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.

2 [Foundational] • Target: Critical points and extrema

Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.



3 [Foundational] • Target: The Extreme Value Theorem

State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.

4 [Foundational] • Target: Rolle's Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.

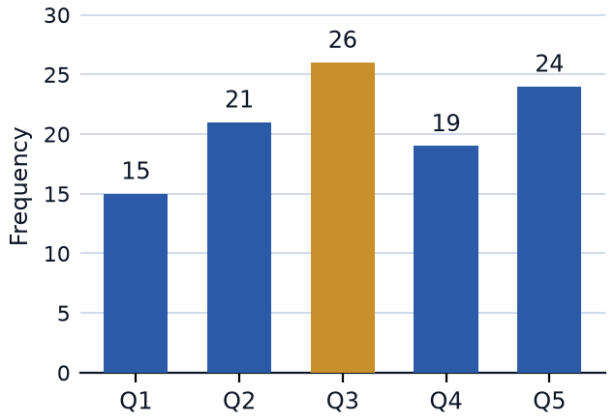
5 [Foundational] • Target: Rolle's Theorem

Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.

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6 [Foundational] • Target: The Mean Value Theorem

For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.



7 [Foundational] • Target: The Mean Value Theorem

A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using the MVT to bound function values

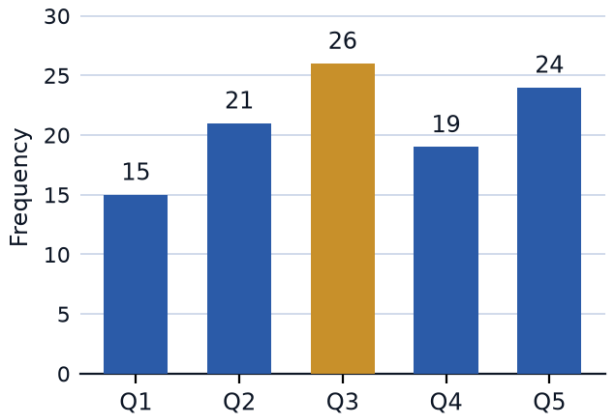
Given $f(2) = 5$ and $-3 \le f'(x) \le 4$ for all x, use the MVT to find the range of possible values for $f(6)$.

9 [Intermediate] • Target: Consequences of the MVT: monotonicity

Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.

10 [Intermediate] • Target: Consequences of the MVT: monotonicity

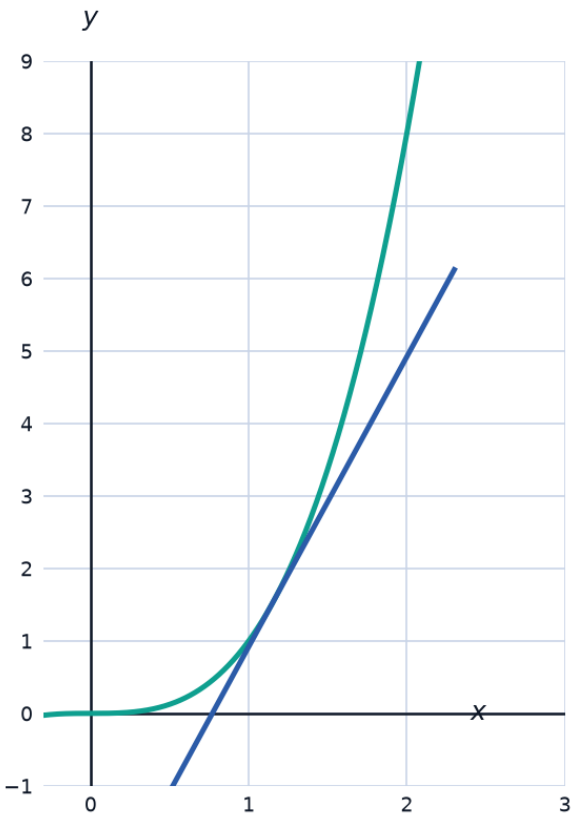
Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.



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11 **[Intermediate]** • **Target: Visualizing the Mean Value Theorem**

The graph shows $f(x) = x^3$ on $[0, 2]$ together with the secant line through the endpoints and the tangent line at $x = \frac{2}{\sqrt{3}}$ (found earlier).



12 **[Intermediate]** • **Target: Absolute extrema on a closed interval — a capstone problem**

Find the absolute maximum and minimum of $f(x) = x^4 - 4x^2 + 3$ on $[-2, 2]$, checking all critical points and endpoints.

13 **[Intermediate]** • **Target: Rolle's Theorem with trigonometric functions**

Verify that Rolle's Theorem applies to $f(x) = \sin x$ on $[0, \pi]$, and find the value of c guaranteed by the theorem.

14 **[Intermediate]** • **Target: An applied Mean Value Theorem problem**

A runner completes a 10 km race in 40 minutes. Using the Mean Value Theorem, find the runner's average speed, and explain why there must be some instant during the race when the runner's instantaneous speed equals this average.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

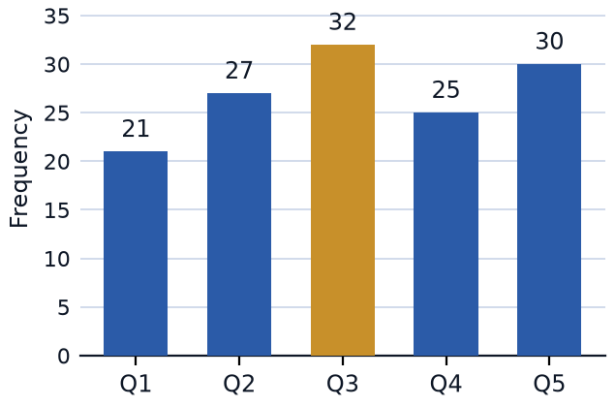
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15 [Challenging] • Target: The Extreme Value Theorem • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Given $f(1) = -5$ and $-2 \le f'(x) \le 2$ for all x, use the MVT to find the largest possible value of $f(2)$.

16 [Challenging] • Target: Rolle's Theorem • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Given $f(-3) = 3$ and $-3 \le f'(x) \le 5$ for all x, use the MVT to find the largest possible value of $f(-1)$.



17 [Challenging] • Target: The Mean Value Theorem • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Given $f(-4) = -3$ and $-6 \le f'(x) \le 2$ for all x, use the MVT to find the largest possible value of $f(2)$.

18 [Challenging] • Target: Using the MVT to bound function values • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Given $f(1) = 1$ and $-2 \le f'(x) \le 5$ for all x, use the MVT to find the largest possible value of $f(4)$.

19 [Challenging] • Target: Consequences of the MVT: monotonicity • Cross-Lesson: 8.01 Area Between Curves

Given $f(3) = 5$ and $-2 \le f'(x) \le 3$ for all x, use the MVT to find the largest possible value of $f(6)$.

20 [Challenging] • Target: Visualizing the Mean Value Theorem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Find the larger critical point of $f(x) = x^3 - 9x^2 + 24x$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Critical points and extrema

Verify that Rolle's Theorem applies to $f(x) = x^2 + 4x$ on $[-6, 2]$, and find the value of c guaranteed by the theorem.

a. -2

b. -6

c. -1

d. 2

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22 [Foundational] • Target: The Extreme Value Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2$ on $[-2, 2]$, and find the value of c guaranteed by the theorem.

a. 1

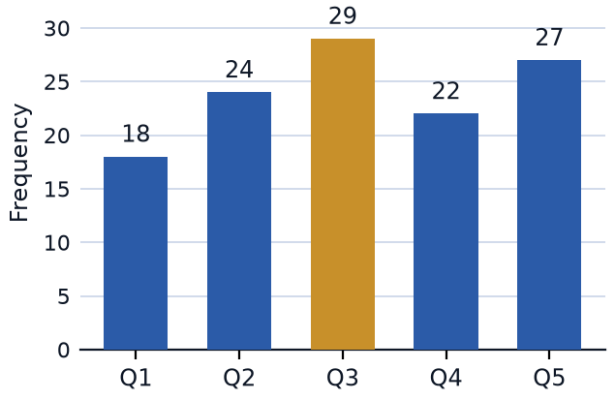
b. 0

c. 2

d. -2

23 [Foundational] • Target: Rolle's Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 10x$ on $[0, 10]$, and find the value of c guaranteed by the theorem.



a. 5

b. 10

c. 6

d. 0

24 [Foundational] • Target: The Mean Value Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 + 4x$ on $[-5, 1]$, and find the value of c guaranteed by the theorem.

a. -1

b. -2

c. -5

d. 1

25 [Foundational] • Target: Using the MVT to bound function values

Verify that Rolle's Theorem applies to $f(x) = x^2 + 12x$ on $[-9, -3]$, and find the value of c guaranteed by the theorem.

a. -5

b. -6

c. -3

d. -9

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26 [Foundational] • Target: Consequences of the MVT: monotonicity

Verify that Rolle's Theorem applies to $f(x) = x^2 + 10x$ on $[-6, -4]$, and find the value of c guaranteed by the theorem.

a. -3

b. -6

c. -4

d. -5

27 [Foundational] • Target: Visualizing the Mean Value Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 6x$ on $[2, 4]$, and find the value of c guaranteed by the theorem.

a. 4

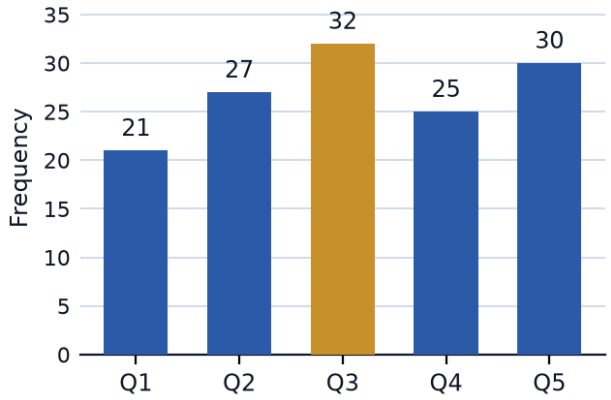
b. 5

c. 2

d. 3

28 [Foundational] • Target: Absolute extrema on a closed interval — a capstone problem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 12x$ on $[1, 11]$, and find the value of c guaranteed by the theorem.



a. 6

b. 1

c. 7

d. 11

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Rolle's Theorem with trigonometric functions

Given $f(1) = -5$ and $-4 \le f'(x) \le 4$ for all x, use the MVT to find the largest possible value of $f(7)$.

a. -29

b. 25

c. 15

d. 19

30 [Intermediate] • Target: An applied Mean Value Theorem problem

Given $f(-3) = 6$ and $-4 \le f'(x) \le 5$ for all x, use the MVT to find the largest possible value of $f(5)$.

a. 41

b. 46

c. -26

d. 54

31 [Intermediate] • Target: Critical Points & First Derivative Test

Given $f(2) = 1$ and $-5 \le f'(x) \le 4$ for all x, use the MVT to find the largest possible value of $f(9)$.

a. -34

b. 36

c. 25

d. 29

32 [Intermediate] • Target: Mean Value Theorem (MVT)

Given $f(1) = -7$ and $-3 \le f'(x) \le 6$ for all x, use the MVT to find the largest possible value of $f(9)$.

a. 41

b. 35

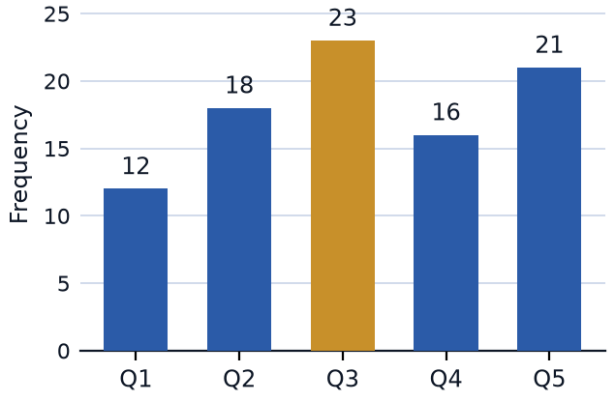
c. 49

d. -31

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33 [Intermediate] • Target: Absolute Extrema on Closed Intervals (Candidates Test)

Given $f(4) = 8$ and $-3 \le f'(x) \le 4$ for all x, use the MVT to find the largest possible value of $f(6)$.



a. 12

b. 18

c. 16

d. 2

34 [Intermediate] • Target: Critical points and extrema

Find the larger critical point of $f(x) = x^3 + 6x^2 - 36x$.

a. -4

b. 2

c. -6

d. -12

35 [Intermediate] • Target: The Extreme Value Theorem

Find the larger critical point of $f(x) = x^3 + 9x^2$.

a. -2

b. 3

c. -6

d. 0

36 [Intermediate] • Target: Rolle's Theorem

Given $-5 \le f'(x) \le 6$ for all x, use the MVT to find the width of the range of possible values for $f(1) - f(0)$.

a. 11

b. 1

c. 14

d. 12

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37 **[Intermediate]** • **Target: The Mean Value Theorem**

Given $-2 \le f'(x) \le 1$ for all x, use the MVT to find the width of the range of possible values for $f(2) - f(-2)$.

a. 3

b. 12

c. 15

d. 16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Using the MVT to bound function values** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Does Rolle's Theorem guarantee a $c \in (-4, 6)$ with $f'(c) = 0$ for $f(x) = (x - 1)^2$ on $[-4, 6]$?

a. Yes, but only because f is increasing on the whole interval.

b. Yes — $f(-4) = f(6) = 25$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = 1$.

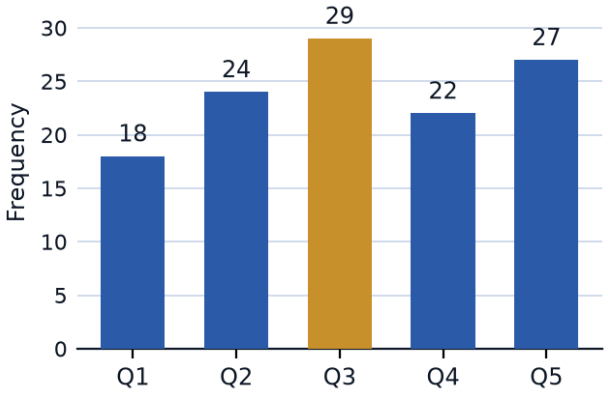
c. No — although $f(-4) = f(6)$, f is not differentiable on $(-4, 6)$, so Rolle's Theorem does not apply.

d. No — Rolle's Theorem never applies to functions of the form $(x - 1)^2$.

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39 [Challenging] • Target: Consequences of the MVT: monotonicity • Cross-Lesson: 8.01 Area Between Curves

Does Rolle's Theorem guarantee a $c \in (-7, 5)$ with $f'(c) = 0$ for $f(x) = (x + 1)^2$ on $[-7, 5]$?



a. Yes, but only because f is increasing on the whole interval.

b. No — Rolle's Theorem never applies to functions of the form $(x + 1)^2$.

c. Yes — $f(-7) = f(5) = 36$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = -1$.

d. No — although $f(-7) = f(5)$, f is not differentiable on $(-7, 5)$, so Rolle's Theorem does not apply.

40 [Challenging] • Target: Visualizing the Mean Value Theorem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Does Rolle's Theorem guarantee a $c \in (2, 6)$ with $f'(c) = 0$ for $f(x) = |x - 4|$ on $[2, 6]$?

a. No — Rolle's Theorem never applies to absolute value functions, regardless of the interval.

b. Yes, since $f(2) = f(6)$ is the only hypothesis Rolle's Theorem requires.

c. No — although $f(2) = f(6) = 2$, f has a corner at $x = 4 \in (2, 6)$ and is not differentiable there, so one hypothesis of Rolle's Theorem fails.

d. Yes — $f(2) = f(6) = 2$, and f is continuous and differentiable everywhere, so Rolle's Theorem applies; here $c = 4$.

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41 [Challenging] • Target: Critical Points & First Derivative Test • Cross-Lesson: 1.01 Limits Graphically and Numerically

By the First Derivative Test, what type of extremum does $f(x) = x^3 - 3x^2 - 9x + 5$ have at $x = -1$?

a. Local maximum because f' changes from positive to negative

b. Local minimum because f' changes from negative to positive

c. Inflection point

d. No extremum

42 [Challenging] • Target: Mean Value Theorem (MVT) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

What are the conditions required to apply the Mean Value Theorem on $[a, b]$?

a. Continuous on $[a, b]$ and differentiable on (a, b)

b. Differentiable on $[a, b]$ and $f(a) = f(b)$

c. Continuous on (a, b) only

d. $f'(c) = 0$ on $[a, b]$

43 [Challenging] • Target: Mean Value Theorem (MVT) • Cross-Lesson: 2.01 The Definition of the Derivative

Why does the MVT NOT apply to $f(x) = |x|$ on $[-2, 2]$?

a. f is not differentiable at $x = 0$

b. f is not continuous on $[-2, 2]$

c. $f(-2) \neq f(2)$

d. The interval is not symmetric

44 [Challenging] • Target: Rolle's Theorem • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Verify Rolle's Theorem for $f(x) = x^2 - 5x + 4$ on $[1, 4]$. What is the value of c?

a. 2.5

b. 2.0

c. 3.0

d. 1.5

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45 [Challenging] • Target: Absolute Extrema on Closed Intervals (Candidates Test) • Cross-Lesson: 3.01 The Chain Rule

Find the absolute maximum value of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on $[-2, 3]$.

a. 8

b. 1

c. -19

d. -8

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